

Project 1 — Exploratory Data Analysis (EDA)

Overview

This project demonstrates core exploratory data analysis skills using Python. The goal is to take a raw dataset, clean it, explore patterns, and communicate insights through visualizations.

Tools Used

- Python
- pandas
- NumPy
- Matplotlib
- Plotly Express

Files

- eda.ipynb — main notebook
- CPS 2.1 Removals - by Region with Child Demographics FY2016-2025.xlsx — dataset used for analysis

Key Steps

- Load and inspect the dataset
- Handle missing values
- Clean and normalize columns
- Generate descriptive statistics
- Explore trends and correlations
- Visualize findings
- Summarize insights

Questions Answered

- What are the main trends in the dataset?
- Which variables are most correlated?
- Are there any outliers or anomalies?
- What insights can be drawn from the visualizations?

Univariate Analysis

A simple statistical data analysis that examines each variable (var) independently to understand its value distribution. This section explores each variable independently to understand its distribution, central tendencies, and overall behavior. Univariate analysis helps identify patterns, skewness, outliers, and category imbalances before moving into deeper multivariate relationships.